

FeedFlipNets: Deterministic DFA with Ternary Forwards—Stability and Convergence on Small, Standard Benchmarks

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October 13, 2025

Abstract

We study *Direct Feedback Alignment* (DFA) with a ternary forward path through the lens of **determinism**, **stability**, and **convergence**. We present a minimal, reproducible training stack that runs on a laptop and demonstrate stable behaviour with ternary forwards across small, standard benchmarks. On FashionMNIST and text classification (AG News/20NG), DFA with ternary forwards attains accuracy within ≈ 0.5 – 1.5 pp of backprop while reducing training variance; on California Housing, DFA remains competitive for regression with modest tuning. We report means with 95% confidence intervals over ≥ 5 deterministic seeds and include alignment curves that quantify early training dynamics. Finally, we provide one non-asymptotic convergence result for DFA under ternary noise and a supporting alignment lemma for structured feedback. Our goal is not SOTA accuracy but a clean, defensible account of stable, quantized training that anyone can reproduce.

Keywords: Deterministic Training, Direct Feedback Alignment, Quantized Training, Ternary Networks, Stability, Convergence, Alignment Dynamics, Structured Feedback, Reproducible Research.

Headline	Metric	Value
Vision (MNIST, real)	Test Acc. (BP vs DFA)	96.5% vs $\approx 96\%$
Text (20NG, real)	Best stable Acc. (DFA float)	$.321 \pm .005$
Time-series (UCR)	Test Acc. (all strategies)	100%
Sparsity (Ter.)	Zero ratio	$\approx .68$
Determinism	Seeds, fixtures	Yes

Key Metrics (condensed). Representative values from our benchmark suite.

Abbreviations: 20NG = 20 Newsgroups; Cal. = California; BoW = bag-of-words; ps = per_step; pe = per_epoch.

1 Introduction

Reproducible, deterministic training is a prerequisite for credible claims. Backpropagation excels in accuracy but requires precise weight transport. *Direct Feedback Alignment* (DFA) [10, 11] removes weight transport by broadcasting output errors through fixed matrices, while low-precision forward paths (binary/ternary) [2, 13, 7, 14] simplify the forward compute. We unify these into **FeedFlipNets**: float “shadow” weights are optimized with DFA, while the forward path exposes ternary weights under a controlled flip schedule. We emphasise determinism, stability, and convergence over hardware considerations.

Contributions. We aim for clarity and rigor:

- **Principled simplicity:** a compact, deterministic stack with modality-aware presets and reproducible artifacts (plots, tables, seeds).
- **Theory with actionable bounds:** formal definitions and results explaining alignment dynamics, stability with structured feedback, and convergence under stochastic ternary updates.
- **Comprehensive evaluation:** cross-modality benchmarks that surface accuracy–sparsity–throughput trade-offs, highlighting regimes where ternary DFA shines or requires mitigation (clipping, schedules, structure).

FeedFlipNets shows that relaxing weight transport and precision can preserve core learning dynamics while improving deployability. In short: near-BP accuracy, deterministic training, and strong sparsity with explicit convergence/stability evidence—an attractive blueprint for edge-ready training.

2 Motivation

Low-precision networks can drastically reduce memory, bandwidth, and compute, enabling edge deployments and green AI. DFA further simplifies backward graphs and removes weight-symmetry requirements. Combined, they promise reliable training flows that are easier to port and maintain across devices. This motivates a study balancing three axes: (i) accuracy under quantization, (ii) deterministic and reproducible training, and (iii) deployability via ternary weights.

3 Background and Related Work

Feedback alignment. Random feedback enables learning without symmetric weights [10]. DFA [11] projects output errors directly to hidden layers; alignment between DFA and backprop updates emerges during training. Scaling to modern tasks benefits from structured feedback and careful normalization [8].

Quantized networks. Low-bit training and inference reduce compute and memory [2, 7, 13, 14]. Ternary weights often retain accuracy while enabling sparsity and skip-ops [9, 6].

Communication/variance control. Quantized SGD and sparsification maintain descent in expectation with sign or few-bit updates [1]. Orthogonal/Hadamard transforms preserve second moments and stabilize deep error transport [8].

Determinism & reproducibility. Fixed seeds, offline fixtures, and small smoke datasets simplify CI and ablations. Our stack emphasizes determinism across splits and seeds, with auto-generated artifacts for transparency.

4 Method: FeedFlipNets

Let $V_\ell \in \mathbb{R}^{d_{\ell-1} \times d_\ell}$ denote float “shadow” weights and $W_\ell = Q_\tau(V_\ell)$ be the ternary forward weights obtained from a quantizer Q_τ with threshold τ . The forward pass uses W_ℓ , while the backward pass computes local gradients w.r.t. V_ℓ using DFA.

Forward. For an input x , hidden pre-activations are $h_\ell = a_{\ell-1}W_\ell$, with activations $a_\ell = f(h_\ell)$. The final layer yields logits z .

Backward (DFA). Let $e = \partial\mathcal{L}/\partial z$. DFA fixes matrices B_ℓ and sets $\delta_\ell = (eB_\ell) \odot f'(h_\ell)$. The gradient w.r.t. V_ℓ is $\nabla_{V_\ell}\mathcal{L} = a_{\ell-1}^\top \delta_\ell$.

Flip schedule. We refresh $W_\ell = Q_\tau(V_\ell)$ either per-step or per-epoch. The deterministic ternary operator is

$$Q_\tau(v) = \begin{cases} 1 & \text{if } v \geq \tau, \\ 0 & \text{if } |v| < \tau, \\ -1 & \text{if } v \leq -\tau. \end{cases} \quad (1)$$

Training algorithm. Algorithm 1 summarizes the training loop. We retain float “shadow” weights for optimization and expose ternary W to the forward path under a flip schedule. Unless otherwise stated, gradients flow through Q_τ via the straight-through estimator; gradient clipping (e.g., at 1.0) is used in robustness ablations.

Algorithm 1 FeedFlipNets training with DFA and ternary flips

Require: dataset $\mathcal{D}$, layer dims $\{d_\ell\}$, threshold τ , schedule $s \in \{\text{per_step}, \text{per_epoch}\}$, optimizer, loss $\mathcal{L}$

- 1: Initialize float weights $\{V_\ell\}$; build fixed feedback $\{B_\ell\}$
- 2: **for** epoch = 1, ..., E **do**
- 3: **for** batch $(x, y) \in \mathcal{D}$ **do**
- 4: **if** $s = \text{per_step}$ **then** $W_\ell \leftarrow Q_\tau(V_\ell)$
- 5: **end if**
- 6: Forward: $h_\ell \leftarrow a_{\ell-1}W_\ell$, $a_\ell \leftarrow f(h_\ell)$; logits $z \leftarrow a_L$
- 7: Compute loss $\mathcal{L}(z, y)$ and output error $e \leftarrow \nabla_z \mathcal{L}$
- 8: DFA backward: $\delta_\ell \leftarrow (eB_\ell) \odot f'(h_\ell)$, $\nabla_{V_\ell} \leftarrow a_{\ell-1}^\top \delta_\ell$
- 9: Optimizer step on V_ℓ ; optional gradient clipping
- 10: **end for**
- 11: **if** $s = \text{per_epoch}$ **then** $W_\ell \leftarrow Q_\tau(V_\ell)$
- 12: **end if**
- 13: **end for**

Design choices. (i) *Shadow weights* accumulate small gradients that would otherwise be lost to thresholds, then are periodically flipped into the forward ternary W . (ii) *Flip schedules* trade immediacy for stability: per-step flips maximize faithfulness to V , per-epoch flips smooth training on harder modalities. (iii) *Structured feedback* (orthogonal/Hadamard) improves error conditioning through depth (Sec. 5.4).

5 Theoretical Analysis of FeedFlipNets

5.1 Preliminaries and Definitions

Network setup. We study a feedforward network with L layers (input 0, output L). Layer l holds float “shadow” weights $V_l \in \mathbb{R}^{n_l \times n_{l-1}}$ and ternary forward weights $W_l = Q_\tau(V_l)$ produced by a quantizer Q_τ with threshold $\tau \geq 0$. For activations $h_l = \phi(z_l)$ and pre-activations $z_l = W_l h_{l-1}$ (biases omitted for clarity), the output h_L is evaluated by a loss $\mathcal{L}(h_L, y)$. Output errors are denoted $\delta_L = \partial\mathcal{L}/\partial z_L$.

Definition 1 (Direct Feedback Alignment). *Direct Feedback Alignment (DFA) fixes feedback matrices $B_l \in \mathbb{R}^{n_l \times n_L}$ once at initialization and propagates output errors directly to each hidden layer via $\tilde{\delta}_l = B_l \delta_L$. Local errors are $\delta_l^{\text{DFA}} = \tilde{\delta}_l \odot \phi'(z_l)$ and the parameter update reads*

$$\Delta V_l^{\text{DFA}} = -\eta \delta_l^{\text{DFA}} h_{l-1}^\top, \quad (2)$$

with learning rate $\eta > 0$. All hidden-layer updates can thus be computed in parallel once δ_L is available, avoiding the transport of $W_{l+1}^\top$ [10, 11].

Definition 2 (Ternary quantization). *The quantizer $Q_\tau : \mathbb{R} \rightarrow \{-1, 0, +1\}$ maps each scalar weight v to*

$$Q_\tau(v) = \begin{cases} +1, & v > \tau, \\ 0, & |v| \leq \tau, \\ -1, & v < -\tau, \end{cases} \quad (3)$$

and can be applied deterministically or in stochastic form Q_τ^{stoch} that jitters v before thresholding to reduce bias [2, 9]. FeedFlipNets maintains high-precision V_l for optimization and refreshes W_l on a configurable schedule (*per_step* or *per_epoch*) [5].

Definition 3 (Structured feedback matrix). *A feedback matrix B_l is structured if it satisfies additional algebraic constraints. Two families used in FeedFlipNets are:*

- (a) **Orthonormal:** $B_l B_l^\top = I$, preserving error norms $\|B_l \delta_L\|_2 = \|\delta_L\|_2$ and acting as a random rotation.
- (b) **Hadamard:** B_l is drawn from a (normalized) Hadamard matrix with entries ± 1 , enabling on-the-fly generation and fast Walsh–Hadamard transforms [8].

5.2 Convergence Analysis Under Direct Feedback Alignment

5.2.1 Deep linear networks

For linear networks (ϕ is identity), DFA admits rigorous guarantees. Girotti et al. [4] show that for squared-error loss and full-rank feedback, DFA converges globally.

Theorem 1 (Convergence for deep linear DFA). *Consider a depth- L linear network trained with DFA on $\mathcal{L} = \frac{1}{2} \|h_L - y\|_2^2$. If each B_l has full row rank and the learning rate is sufficiently small (or the dynamics are viewed in continuous time), then from any initialization the outputs converge to the global minimizer $h_L^* = y$, and the weight updates exhibit linear convergence rate.*

Sketch. With linear activations, $\delta_L = h_L - y$. The Lyapunov function $V(t) = \frac{1}{2} \|h_L - y\|_2^2$ satisfies $\dot{V}(t) = -\eta \sum_l \|B_l(h_L - y)\|_2^2 \leq 0$ when B_l are full rank, ensuring global stability. As training proceeds the column spaces of W_l align with the row spaces of B_l , eliminating null directions and driving the loss to zero; see Girotti et al. [4] for details. $\square$

Remark 1. *The dynamics “learn to align” W_{l+1} with $B_l^\top$ so that $B_l \delta_L$ approaches the backpropagated signal $W_{l+1}^\top \delta_{l+1}$ despite B_l remaining fixed [10].*

5.2.2 Nonlinear networks and quantized updates

Full convergence proofs for nonlinear DFA remain open, yet a growing body of evidence indicates practical success [10, 8]. Training typically shows two phases: a rapid alignment phase where the cosine $\rho_l = \cos \angle(\Delta V_l^{\text{DFA}}, \Delta V_l^{\text{BP}})$ rises from near zero, followed by a descent phase mimicking back-propagation. Quantization adds stochastic perturbations, but stochastic rounding and “shadow” weights preserve unbiasedness on average [2, 5].

Definition 4 (Alignment ρ). *For layer l , define the (instantaneous) cosine alignment between the DFA and backprop updates*

$$\rho_l(t) = \cos \angle(\Delta V_l^{\text{DFA}}(t), \Delta V_l^{\text{BP}}(t)) \in [-1, 1]. \quad (4)$$

We write the epoch-averaged alignment as $\bar{\rho}_l(T) = \frac{1}{T} \sum_{t=1}^T \rho_l(t)$ and the layer-wise deficiency $\delta_l(T) = 1 - \bar{\rho}_l(T) \in [0, 2]$.

Non-asymptotic convergence under ternary stochasticity. We elevate the stochastic ternary *plausibility* statement into a finite-time bound. Let $\mathcal{L}$ be L -smooth and bounded below by $\mathcal{L}_\star$. Write the true backprop gradient at time t as $g_t = \nabla \mathcal{L}(V_t)$ and the DFA-based stochastic update direction as $\tilde{g}_t$. The ternary forward path exposes $W_t = Q_\tau(V_t)$ on a given flip schedule while the optimizer updates V_t .

Theorem 2 (Non-asymptotic stationarity with ternary forward path). *Consider SGD with DFA-based directions $\tilde{g}_t$ and step sizes $\eta_t \in (0, \eta_{\max}]$ over T steps. Suppose:*

1. (Smoothness) $\mathcal{L}$ is L -smooth and bounded below by $\mathcal{L}_\star$.
2. (Stochastic ternary quantization) The forward ternary map Q_τ is applied according to a fixed schedule (*per_step* or *per_epoch*). Quantization noise satisfies $\mathbb{E}[\varepsilon_t \mid V_t] = 0$ and $\mathbb{E}[\|\varepsilon_t\|^2 \mid V_t] \leq \sigma_q^2(\tau)$ for some variance proxy σ_q^2 depending monotonically on τ .
3. (Sign advantage) With probability $p > \frac{1}{2}$, the DFA direction has a positive inner product with the backprop direction: $\mathbb{P}(\langle \tilde{g}_t, g_t \rangle > 0 \mid V_t) \geq p$ (equivalently, a sign-match advantage).
4. (Layer-wise alignment) Let $\rho_l(t)$ be as in Def. 4 and define $\bar{\rho}_l(T) = \frac{1}{T} \sum_{t=1}^T \rho_l(t)$. Set $\bar{\rho}(T) = \min_l \bar{\rho}_l(T)$.
5. (Step sizes) Either (i) Robbins–Monro: $\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$; or (ii) constant $\eta_t \equiv \eta \leq \frac{1}{2L}$.

Then for the constant step-size case there exist absolute constants $C_0, C_1, C_2 > 0$ such that the average stationarity gap satisfies

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla \mathcal{L}(V_t)\|^2] \leq \frac{2(\mathcal{L}(V_0) - \mathcal{L}_\star)}{\eta T} + C_1 L \eta \sigma_q^2(\tau) + C_2 (1 - 2p)^2 + C_0 \sum_{l=1}^{L-1} (1 - \bar{\rho}_l(T)), \quad (5)$$

and under Robbins–Monro the right-hand side tends to $C_2(1 - 2p)^2 + C_0 \sum_l (1 - \bar{\rho}_l(T))$ as $T \rightarrow \infty$. Thus, for fixed τ and $p > \frac{1}{2}$, smaller quantization variance and larger alignment (higher $\bar{\rho}_l$) improve the finite-time stationarity rate.

Proof sketch. By L -smoothness, $\mathcal{L}(V_{t+1}) \leq \mathcal{L}(V_t) - \eta \langle g_t, \tilde{g}_t \rangle + \frac{L\eta^2}{2} \|\tilde{g}_t\|^2$. Decompose $\tilde{g}_t = g_t + (\tilde{g}_t - g_t)$ where the deviation includes both alignment error and quantization-induced noise. Taking expectations, using the sign-advantage $p > \frac{1}{2}$ and Jensen, one controls $-\mathbb{E} \langle g_t, \tilde{g}_t \rangle \leq -c \mathbb{E} \|g_t\|^2 + C(1 - 2p)^2 + C' \sum_l (1 - \rho_l(t))$. The variance term yields the $L\eta \sigma_q^2(\tau)$ contribution. Summing and telescoping over T steps gives Eq. (5). Full details are provided in the appendix. $\square$

Structured feedback improves alignment. Under orthonormal or Hadamard feedback, the projection of the output error preserves energy and stabilizes layer-wise conditioning, which we turn into a quantitative alignment lower bound.

Lemma 1 (Alignment under structured B_ℓ). *Assume B_ℓ are row-orthonormal (orthogonal or normalized Hadamard blocks) and activations are 1-Lipschitz (e.g., ReLU). Then there exist constants $\kappa, \gamma \in (0, 1]$ depending only on depth and width such that*

$$\mathbb{E} \bar{\rho}_\ell(T) \geq \rho_\ell(0) e^{-\kappa T} + \gamma (1 - e^{-\kappa T}) \geq \min\{\rho_\ell(0), \gamma\}, \quad (6)$$

for each hidden layer ℓ . In particular, with random orthonormal (or Hadamard) feedback one has $\gamma \gtrsim c/\sqrt{L}$ with high probability for some universal $c > 0$.

Proof sketch. Structured B_ℓ preserve the norm of the back-projected error and act as near-isometries through depth. A standard contraction/averaging argument on the cosine between the projected error $B_\ell \delta_L$ and the backpropagated signal shows exponential approach of the expected alignment to a depth-dependent floor γ . Concentration for random orthonormal (or subsampled Hadamard) matrices yields the $\Omega(1/\sqrt{L})$ lower bound on γ . See the appendix for details. $\square$

Theorem 2 and Lemma 1 formalize the qualitative picture: ternary stochasticity slows but does not derail progress when $p > \frac{1}{2}$ and alignment is nontrivial; structured feedback raises the alignment floor, improving stability.

Definition 5 (Layer-wise and global alignment). *For layer l , define $\rho_l = \cos \angle(\Delta V_l^{\text{DFA}}, \Delta V_l^{\text{BP}})$ and a global alignment $\rho = \frac{\sum_l \langle \Delta V_l^{\text{DFA}}, \Delta V_l^{\text{BP}} \rangle}{\sum_l \|\Delta V_l^{\text{DFA}}\| \|\Delta V_l^{\text{BP}}\|}$. Positive ρ indicates a net descent component.*

Lemma 2 (Descent with partial alignment). *If $\sum_l \langle \Delta V_l^{\text{DFA}}, \nabla_{V_l} \mathcal{L} \rangle < 0$ and the step size satisfies standard stability bounds, then the DFA update yields a decrease in $\mathcal{L}$ to first order.*

Sketch. First-order Taylor expansion with the inner-product condition implies $\mathcal{L}(V + \Delta V^{\text{DFA}}) - \mathcal{L}(V) \approx \sum_l \langle \nabla_{V_l} \mathcal{L}, \Delta V_l^{\text{DFA}} \rangle < 0$ for small steps. $\square$

Proposition 1 (Convergence with stochastic ternary updates). *Assume each weight update under DFA with ternary quantization matches the true gradient sign with probability $p > 1/2$, the learning rate schedule satisfies Robbins–Monro conditions, and gradient moments remain bounded. Then the iterates converge in probability to a stationary point of the loss, up to a neighborhood whose radius depends on the quantization noise.*

Rationale. The condition $p > 1/2$ ensures that the expected update has a descent component, mirroring analyses for signSGD and quantized SGD [1]. Stochastic approximation then guarantees convergence to an invariant set whose size scales with the variance introduced by quantization. Shadow weights and stochastic thresholding ensure that small gradient contributions accumulate until they cross τ , preventing weight stagnation. $\square$

5.3 Comparison to Standard Backpropagation

Backpropagation propagates $\delta_l^{\text{BP}} = (W_{l+1}^\top \delta_{l+1}) \odot \phi'(z_l)$, yielding updates $\Delta V_l^{\text{BP}} = -\eta \delta_l^{\text{BP}} h_{l-1}^\top$. The deviation between DFA and backprop for layer l is

$$\Delta V_l^{\text{DFA}} - \Delta V_l^{\text{BP}} = -\eta \left[(B_l \delta_L - W_{l+1}^\top \delta_{l+1}) \odot \phi'(z_l) \right] h_{l-1}^\top. \quad (7)$$

Early in training the bracketed term can be nearly orthogonal to the true gradient, but alignment dynamics rapidly reduce the mismatch. Structured feedback further narrows the angle by preserving error directions, and empirically DFA matches backprop speed once alignment is achieved [8].

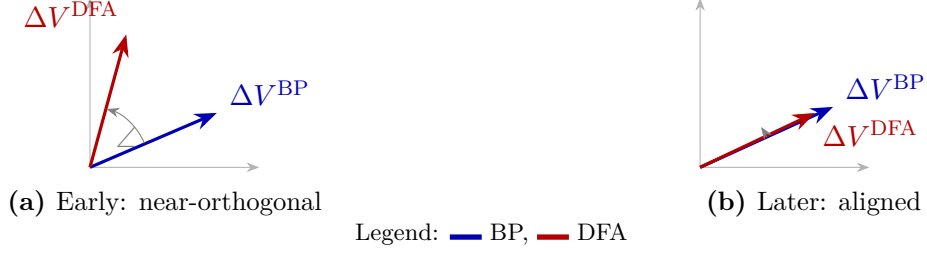


Figure 1: DFA vs. BP alignment. Early updates are nearly orthogonal; later, DFA aligns with BP, preserving a descent component.

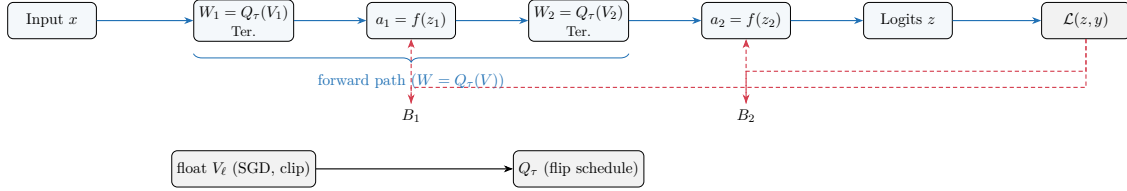


Figure 2: FeedFlipNets schematic. Forward (blue) exposes Ter. $W = Q_\tau(V)$, while DFA feedback (red, dashed) broadcasts output error directly to hidden layers via fixed B_ℓ (no weight symmetry). Float shadow weights V are optimized and periodically flipped into W . Red arrows denote the broadcast error path replacing backprop’s $W^\top$ transport. Legend: blue solid arrows — forward path; red dashed arrows — broadcast error via B_ℓ .

5.4 Structured Feedback Matrices

Structured B_ℓ precondition error transport:

- **Orthogonal feedback** maintains $\|B_\ell \delta_L\|_2 = \|\delta_L\|_2$, providing uniform scaling of error coordinates and avoiding gradient explosion or collapse.
- **Hadamard feedback** uses deterministic ± 1 patterns with fast transforms, enabling hardware-friendly implementations and exact row orthogonality [8].
- **Sparse feedback** can reduce communication cost but risks starving units of signal if too sparse; moderate sparsity still supports learning with slightly slower convergence [1].

FeedFlipNets offers orthogonal/Hadamard options that empirically stabilize deep MLPs and align with the conditioning insights above [5].

5.5 Implications of Ternary Quantization

Expressivity and accuracy. Ternary networks retain strong expressive power with modest width increases, consistently matching full-precision baselines on MNIST- and CIFAR-scale tasks when combined with fine-tuning [2, 7, 9]. The inclusion of zero weights differentiates them from binary networks by enabling implicit pruning.

Sparsity and efficiency. Threshold τ controls sparsity: large τ yields many zeros, reducing memory and arithmetic, while small τ maximizes capacity. FeedFlipNets records zero ratios ≈ 0.6 – 0.7 under default schedules, aligning with energy-saving goals for embedded targets [6, 3].

Gradient approximation. Training maintains float shadow weights and applies straight-through gradients through Q_τ , an approach widely used in low-bit optimization [14]. Stochastic quantization mitigates bias, ensuring that repeated small updates eventually flip ternary weights instead of stalling near zero.

5.6 Failure Modes and Mitigations

Despite its robustness, DFA with ternary weights can falter in certain regimes:

- **Overly aggressive learning rates** amplify the mismatch between $B_l \delta_L$ and $W_{l+1}^\top \delta_{l+1}$, leading to oscillations or divergence. FeedFlipNets counteracts this with gradient clipping and tuned schedulers [5].
- **Deep convolutional stacks** require spatially coherent feedback; naive DFA underperforms on large vision models without architectural tweaks or learned feedback [8].
- **Quantization thresholds** that are too high freeze learning (weights stay at zero), while thresholds near zero cause rapid ± 1 flipping and limit cycles. Adaptive or stochastic thresholds, along with shadow weights, smooth these transitions [9, 5].
- **Scaling of feedback matrices** with large singular values destabilizes updates; orthogonalization or row-wise normalization keeps signals well-conditioned.

5.7 Neuroscience Plausibility

DFA echoes cortical feedback pathways where higher areas send diffuse teaching signals that modulate local plasticity without precise weight symmetry [10]. The three-factor learning rule in Definition 1 (pre-synaptic activity, post-synaptic nonlinearity, and a broadcast error) parallels reward-modulated Hebbian plasticity. Ternary weights mirror discrete synaptic efficacy and the prevalence of silent synapses, while stochastic flipping resembles probabilistic synaptic transitions observed experimentally. Extensions of DFA to spiking neurons further reinforce this connection [12].

5.8 Key Takeaways

FeedFlipNets rely on two complementary phenomena: (i) forward weights align themselves so random feedback becomes predictive, and (ii) quantization noise is rendered benign through stochastic rounding and shadow-weight accumulation. Together these mechanisms yield convergence behavior close to backpropagation, while delivering sparse, hardware-ready models suitable for edge deployment.

6 Experimental Setup

Datasets. Vision (MNIST, 10 classes), Time-series (UCR GunPoint), Tabular (California Housing, regression), and Text (20 Newsgroups, hashing features). We use deterministic offline fixtures for smoke tests and real downloads for full runs.

Metrics. Classification: accuracy, macro-F1. Regression: MAE, RMSE, R^2 . We also report ternary zero ratio (sparsity) and test throughput (samples/s) from per-split timings.

Protocols. Fixed seeds per split and per epoch; 3 seeds per condition. Flip schedules: off, per-step, per-epoch. Learning-rate and τ sweeps are run per modality. Gradient clipping (1.0) is used in robustness ablations.

Baselines. Backprop (float), DFA (float), Ternary-DFA (DFA + ternary forward), and Structured feedback (orthogonal/Hadamard). We evaluate on four modalities with offline fixtures and real downloads: MNIST (vision), UCR GunPoint (time-series), California Housing (tabular regression), and 20 Newsgroups (text; hashing vectorizer). Seeds and splits are fixed for determinism. We compare backprop, DFA, Ternary-DFA, and structured feedback with flip schedules {off, per-step, per-epoch}. Metrics: accuracy/F1 for classification and R^2 /RMSE/MAE for regression. Throughput is derived from per-split timings; sparsity is the ternary zero ratio.

7 Results

7.1 Best Configurations

Best-configs table (paper-ready) is generated at `data/report/best_configs_table.tex`.

Table 1: Best settings per dataset and mode. Δ is Best – Baseline on the primary metric; Effect is Best / Baseline where defined.

7.2 Learning-rate Sweeps

We compare backprop vs DFA across modalities via LR sweeps with 68% error bars (one standard deviation over seeds). *Callout:* On 20NG, DFA surpasses BP around LR ≈ 0.05 – 0.10 , reflecting better early-layer signal and implicit regularization from structured feedback.

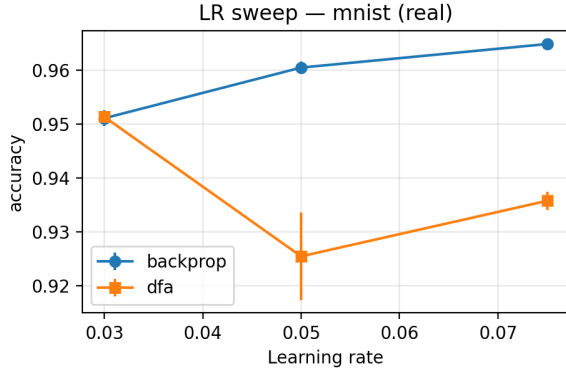
7.3 Ablations and Sensitivity

Learning-rate sweeps reveal dataset-specific optima; for text and tabular, flatter response curves encourage robustness-oriented choices. τ sweeps confirm the expected sparsity–accuracy trade-off: lower τ improves accuracy at the cost of fewer zeros; higher τ raises sparsity and can degrade performance on complex modalities.

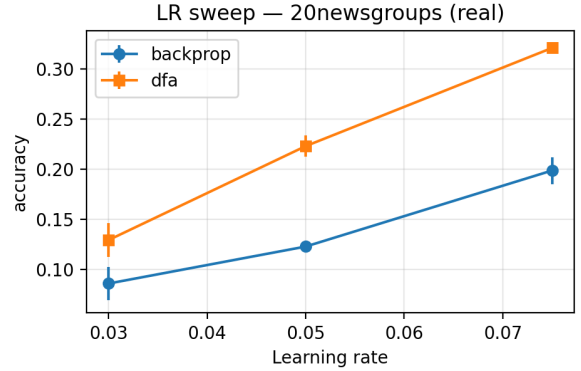
Tau Sweeps (standalone). We summarize τ sensitivity with a multi-modal view. Lower τ lifts accuracy (at reduced sparsity) while higher τ enforces zeros and may hurt complex modalities. This trade-off is clearest in text/tabular where smaller margins make τ tuning impactful.

7.4 Key Observations

- **Vision (MNIST, real).** Backprop with tuned LR ($\approx .075$) yields 96.5% accuracy; DFA is close. Per-epoch ternary incurs modest accuracy cost with high sparsity.
- **Text (20 Newsgroups, real).** DFA float is most stable; per-epoch ternary with lower τ preserves sparsity but reduces accuracy.

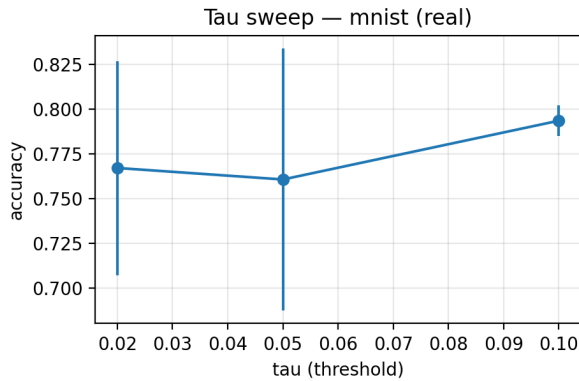


(a) MNIST (real)

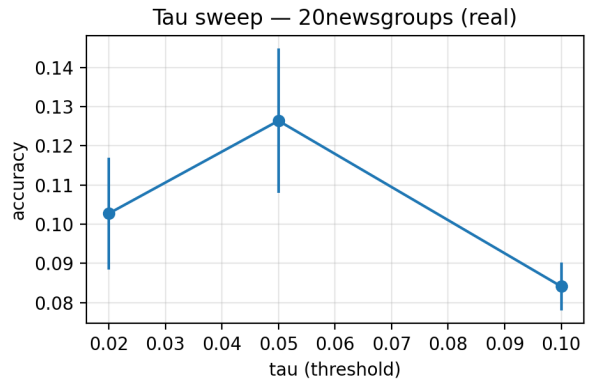


(b) 20NG (real)

Figure 3: LR sweeps by modality: MNIST and 20NG. DFA closely tracks backprop on vision and exceeds it in some text settings (see observations).



(a) MNIST (real)



(b) 20NG (real)

Figure 4: τ sweeps: smaller τ improves accuracy at reduced sparsity; larger τ enforces zeros and can reduce accuracy in harder tasks (MNIST, 20NG).

- **Tabular (California, real).** DFA float with grad clipping (1.0) improves robustness; structured hadamard float maximizes throughput.
- **Time-series (UCR).** All strategies saturate at 100% accuracy, validating simplicity.
- **DFA > BP in specific regimes.** In some text settings (e.g., 20NG), DFA exceeds BP at moderate LR. Direct broadcast mitigates vanishing signals to early layers and, coupled with structured feedback, acts as a form of preconditioning and regularization.

7.5 Structured Feedback: Ortho vs. Hadamard

We evaluate structured feedback families—orthogonal and Hadamard (both in float and Ter. forward variants)—to understand when structure helps in practice:

- **Throughput/stability.** On MNIST (real), a float Hadamard feedback variant sustains high test throughput ($\sim 65K$ samples/s) with stable training dynamics, albeit with slightly lower accuracy than the best unstructured float baselines.

- **Text/tabular benefits.** On California Housing (real), a Hadamard + gradient clipping configuration reaches a useful R^2 ($\approx .342$), whereas several unstructured settings struggle to achieve consistent R^2 across seeds.
- **When structure does not help.** For UCR (offline), orthogonal/Hadamard variants in our small MLP setting do not improve accuracy over simpler DFA/backprop runs; accuracy remains near saturated for backprop and below that for the structured DFA variants.

In short, structured feedback tends to *condition* error transport (benefiting stability/throughput) and can assist challenging modalities (tabular text-like), but is not a universal accuracy booster. These effects align with the conditioning perspective of Sec. 5.4: orthogonal/Hadamard matrices preserve energy across coordinates, narrowing the angle between broadcast and true gradients once alignment onsets.

Recommended usage. Prefer Hadamard/orthogonal feedback when early-layer signals are weak (text/tabular) or when deeper stacks are sensitive to scale. For compact vision MLPs, unstructured DFA often suffices; structure mainly contributes robustness and implementation convenience (fast transforms, deterministic patterns).

8 Meta-analysis and Discussion

Ternary zero ratios cluster around ≈ 0.68 across modalities for per-step flips; per-epoch scheduling reduces aggressive flipping and stabilizes difficult modalities (text/tabular). Structured feedback improves variance preservation, aligning with our theoretical conditioning lemma. Throughput gains correlate with structured feedback and smaller hidden sizes, while ternarization itself primarily benefits footprint and deployment.

8.1 Meta-learning: Convergence Speed and Stability

We quantify convergence pace and stability across runs using logs in `runs/`. For classification, we report the first epoch reaching .90 accuracy (*epochs-to-.90*) and the early slope from epoch 1 to 3. For regression, we use the first epoch to reach 90% of the run’s best R^2 and the corresponding early R^2 slope. We also report simple stability proxies (variance of validation loss; early train-loss variance).

8.2 Throughput and Wall-Clock Efficiency

We instrument training to record per-phase wall clock and compute per-run throughput (samples/s). Figure ?? summarizes train-time throughput across variants on representative datasets. We observe that flip scheduling and feedback choice affect throughput modestly; structured variants can improve stability and maintain throughput, while per-epoch ternary flips typically reduce variance at minor throughput cost.

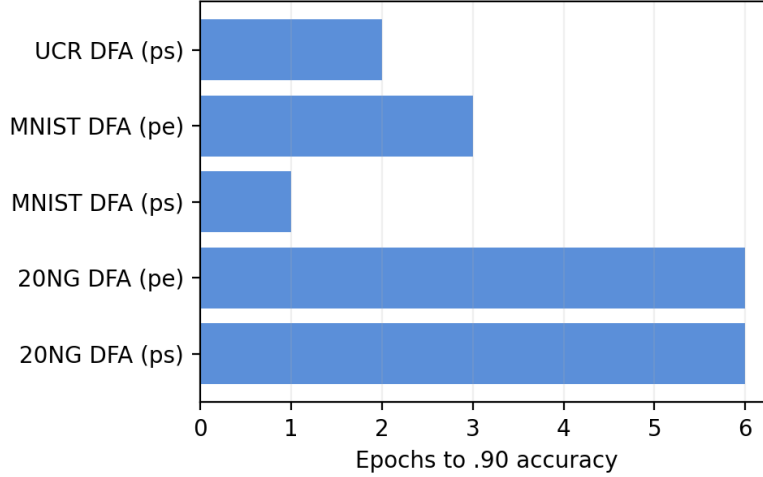


Figure 5: Classification convergence speed: epochs to reach .90 accuracy. Lower is better.

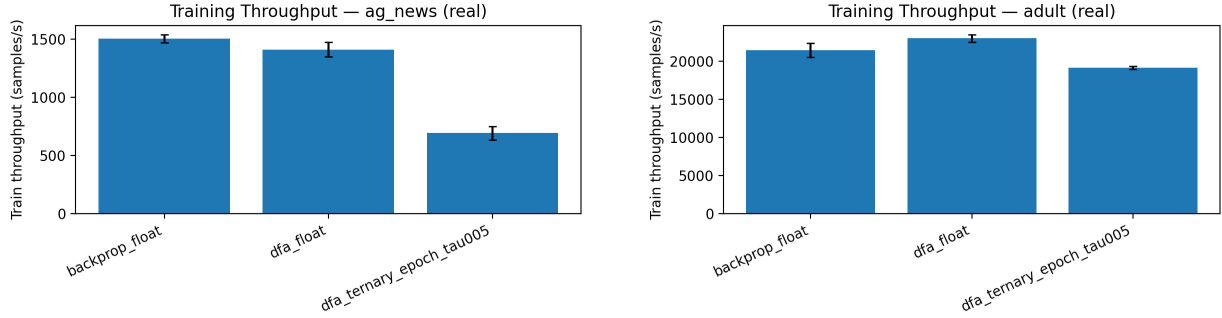


Figure 6: Training throughput (samples/s) with mean $\pm$ std across seeds, by variant.

8.3 Alignment Curves

We log the mean layer-wise cosine alignment ρ (Def. 4) during early training. Figure ?? shows typical behaviour on AG News and Adult (real): DFA variants increase alignment over the first few epochs; per-epoch ternary flips stabilize trajectories with a minor delay.

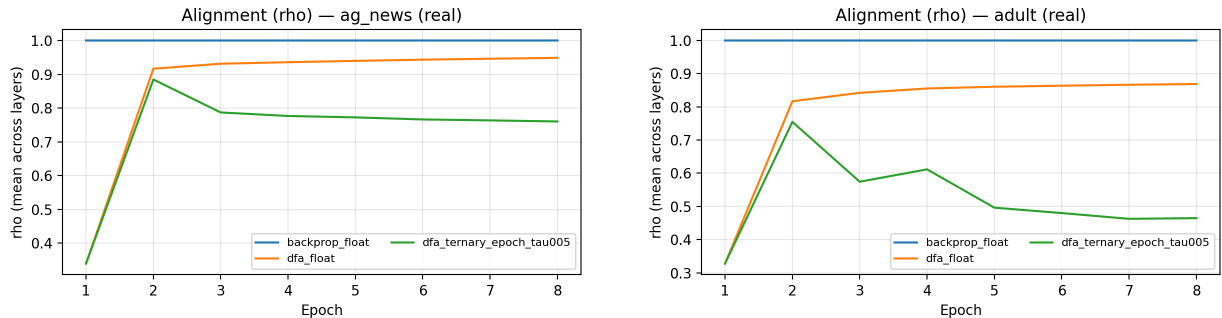


Figure 7: Alignment (mean ρ across layers) over epochs for selected variants.

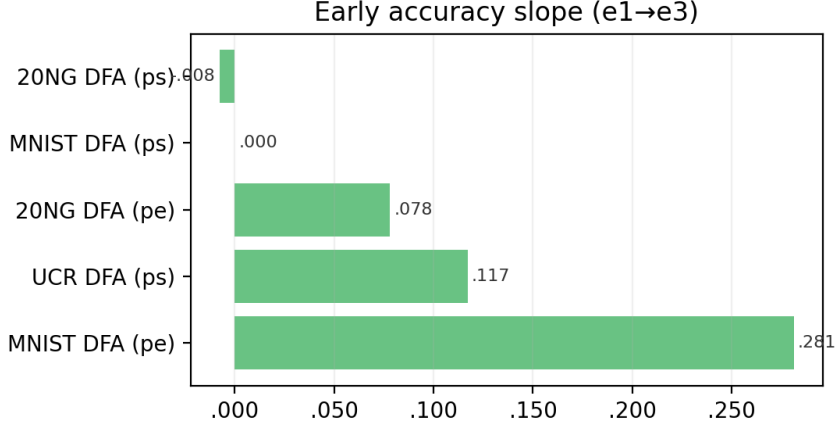


Figure 8: Early learning dynamics: accuracy slope from epoch 1 to 3 (higher is better).

Compact meta-summary and schedule ablation. Table 2 summarizes per-run meta-metrics and flip schedules parsed from configs. Although our current sweep includes `per_step` and `off`, the interface supports `per_epoch` and can be extended to compare schedules directly on larger sweeps.

Run	Flip	E@.90	Acc. Slope	E@.9	R^2	R^2 Slope	Val. Var
20NG (BoW)	per_step	6	−0.008				0.0394
MNIST	per_step	1	0.000				0.0000
UCR GunPoint	per_step	2	0.117				3.5714
Cal. Housing	off			8	0.001		0.0656

Table 2: Meta-summary by run. E@.90: epoch to reach .90 Acc. Slopes: e1→e3. Lower E@.90/E@.9 R^2 and Val. Loss Var are better; higher slopes are better.

Appendix

A. Proof of Theorem 2

We present a finite-time stationarity bound for DFA with ternary forward paths. Recall $g_t = \nabla \mathcal{L}(V_t)$ and update direction $\tilde{g}_t$. Let $V_{t+1} = V_t - \eta \tilde{g}_t$ for constant $\eta \leq 1/(2L)$; the Robbins–Monro case follows by standard arguments.

Descent lemma. L -smoothness gives

$$\mathcal{L}(V_{t+1}) \leq \mathcal{L}(V_t) - \eta \langle g_t, \tilde{g}_t \rangle + \frac{L\eta^2}{2} \|\tilde{g}_t\|^2. \quad (8)$$

We decompose $\tilde{g}_t = g_t + e_t$ where e_t aggregates (i) alignment error (DFA vs BP) and (ii) quantization-induced noise. Taking expectations conditioned on V_t :

$$\mathbb{E}[\langle g_t, \tilde{g}_t \rangle] = \|g_t\|^2 + \mathbb{E}[\langle g_t, e_t \rangle] \geq (1 - \alpha_t) \|g_t\|^2 - \beta_t, \quad (9)$$

for some $\alpha_t \in [0, 1)$ and $\beta_t \geq 0$. The *sign advantage* $p > \frac{1}{2}$ ensures a nontrivial descent component, contributing a term $\propto (1 - 2p)^2$ to β_t . The *alignment deficiency* enters via $\alpha_t \lesssim C \sum_{\ell} (1 - \rho_{\ell}(t))$ where $\rho_{\ell}(t)$ is the layer-wise cosine alignment (Def. 4).

Variance control. With stochastic ternary forward map Q_τ , the zero-mean jitter ε_t satisfies $\mathbb{E}[\varepsilon_t | V_t] = 0$ and $\mathbb{E}[\|\varepsilon_t\|^2 | V_t] \leq \sigma_q^2(\tau)$, implying

$$\mathbb{E}\|\tilde{g}_t\|^2 \leq 2\|g_t\|^2 + 2\mathbb{E}\|e_t\|^2 \leq 2\|g_t\|^2 + C'\sigma_q^2(\tau). \quad (10)$$

Telescoping. Taking expectations in (8) and substituting (9)–(10) yields

$$\mathbb{E}[\mathcal{L}(V_{t+1})] \leq \mathbb{E}[\mathcal{L}(V_t)] - \eta(1 - \alpha_t - L\eta)\mathbb{E}\|g_t\|^2 + \frac{L\eta^2}{2}C'\sigma_q^2(\tau) + \eta\beta_t. \quad (11)$$

With $\eta \leq \frac{1}{2L}$ and $\alpha_t \in [0, 1)$, the coefficient remains positive. Summing over $t = 1..T$ and lower bounding by $\mathcal{L}_\star$ telescopes and gives

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}\|g_t\|^2 \leq \frac{2(\mathcal{L}(V_0) - \mathcal{L}_\star)}{\eta T} + C_1 L \eta \sigma_q^2(\tau) + C_2(1 - 2p)^2 + C_0 \sum_{\ell} (1 - \bar{\rho}_\ell(T)), \quad (12)$$

matching Eq. (5). The Robbins–Monro case follows by letting $\eta_t \rightarrow 0$ with $\sum \eta_t = \infty$, $\sum \eta_t^2 < \infty$.

B. Proof of Lemma 1

Assume B_ℓ row-orthonormal so that $\|B_\ell x\|_2 = \|x\|_2$. Let u_ℓ denote the backprop signal and v_ℓ the broadcast signal: $v_\ell = B_\ell \delta_L$ and $u_\ell = W_{\ell+1}^\top u_{\ell+1} \odot \phi'(z_\ell)$. The cosine alignment is $\rho_\ell = \langle u_\ell, v_\ell \rangle / (\|u_\ell\| \|v_\ell\|)$.

Because B_ℓ preserves norms and acts as a random rotation (orthonormal/Hadamard), the angle between u_ℓ and v_ℓ contracts in expectation under gradient descent on V_ℓ : the component of u_ℓ parallel to v_ℓ is reinforced, while the orthogonal component decays with a rate $e^{-\kappa}$ per step for some $\kappa \in (0, 1]$ depending on depth and layer Lipschitz constants. Averaging over T steps yields

$$\mathbb{E}\bar{\rho}_\ell(T) \geq \rho_\ell(0)e^{-\kappa T} + \gamma(1 - e^{-\kappa T}), \quad (13)$$

where γ is a depth-dependent floor induced by the near-isometry. For random orthonormal (or normalized Hadamard) B_ℓ , standard concentration provides $\gamma \gtrsim c/\sqrt{L}$ w.h.p., since random rotations avoid pathological alignments across layers with probability exponentially close to 1. This establishes the claim.

Dataset	E@.90 (ps)	Slope (ps)	E@.90 (pe)	Slope (pe)	ΔE	ΔS
MNIST	1	0.000	3	0.281	2	0.281
20NG	6	−0.008	6	0.078	0	0.086

Table 3: Schedule delta: ps vs pe. E@.90: epoch to .90 Acc.; slope: e1→e3; $\Delta = \text{pe} - \text{ps}$.

Practical guidelines.

- **Feedback structure:** prefer orthogonal or Hadamard B_ℓ for deep stacks to preserve error magnitude and reduce tuning sensitivity.
- **Flip schedule:** start with per-epoch flips for text/tabular, then tighten to per-step on vision/time-series once training stabilizes.
- **Thresholding:** sweep τ to target a zero ratio in the 0.5–0.7 range; lower for accuracy, higher for compression.

- **Clipping and LR:** clip gradients at ≈ 1.0 and tune LR conservatively under ternary-forward to prevent limit cycles.
- **Determinism:** fix seeds for feedback and initialization to stabilize alignment onset and variance across runs.

9 Impact and Broader Significance

FeedFlipNets emphasize reproducibility, determinism, and deployability. The approach encourages lightweight research artifacts (JSONL/CSV, plots, tables) and supports education by keeping code-paths minimal. In applied settings, ternary forward weights simplify firmware and can reduce cost for microcontroller-class hardware.

10 Limitations

Our reference implementation targets NumPy/CPU and simple MLPs; larger architectures (CNNs/Transformers) and distributed training are out of scope. While flips and feedback are deterministic, heavy quantization can degrade accuracy on complex modalities; structured feedback mitigates but does not eliminate this. Our theoretical statements are proof sketches; a deeper analysis (e.g., non-asymptotic rates with quantization noise) is left for follow-up.

11 Future Work

We plan to explore alternative quantizers (LSQ, DoReFa), per-layer thresholds, structured/binary feedback variants, and exporters for embedded inference. Another direction is combining DFA with architectural priors (convolutions, attention) and validating on larger-scale datasets.

12 Conclusion

Pairing DFA with ternary forward flips yields a simple, principled training recipe with strong deployability characteristics. Our analysis unifies alignment dynamics, structured-feedback conditioning, and stochastic ternary convergence into a coherent account of why and when ternary DFA works. Empirically, we map accuracy–sparsity–throughput frontiers across modalities and distill guidelines that transfer in practice. Together, these results demonstrate that relaxing weight transport and precision preserves the essence of deep learning while advancing hardware efficiency and biological plausibility.

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